Known-Noise Subtraction and Event-Conditioned Complex Masking for Noise Event Removal in Crowd-Sourced Indic Speech

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***Abstract*—Removing transient non-speech noise from crowd-sourced phone recordings must improve both signal fidelity and the accuracy of downstream recognisers. Track 2 of the IndoML 2026 Datathon scores both: SI-SDR on synthetic clips plus the word-error-rate reduction of the mandated SraVaani-1.0 recogniser. Noise events overlap speech at negative local SNR, in-domain training pairs are scarce, and the test generator changed between phases. We propose (i) an event-conditioned complex-mask TF-DPCRN that uses detected noise-event windows as side information, trained on mixtures that reproduce measured test statistics and selected on organiser-grounded held-out data, and (ii) known-noise subtraction: the synthetic test clips reuse noise recordings that the validation release exposes, so matched copies are located by cross-correlation and subtracted, and the enhancer cleans the residual left by their slight alteration. The system ranked 1st on the final leaderboard with Combined 26.4894 (SI-SDR 20.34 dB, ΔWER 6.15 points).**

***Index Terms*—speech enhancement, noise event removal, complex masking, noise cancellation, robust ASR, Vaani**

I. Introduction

Motivation. Speech technology for Indian languages increasingly relies on crowd-sourced corpora such as Vaani [1], recorded on phones in everyday settings. Transient noise events (traffic, animals, music, appliances) degrade recognition and the value of the recordings. Unlike stationary background noise, they are short, diverse and often louder than the speech they overlap, so removing them without damaging the speech is hard.

Task. For each noisy clip (Phase 2 test: 3,178 clips, 10.9 h) a system returns an enhanced waveform of the same length and its transcript by the mandated recogniser SraVaani-1.0 [2]. The score is SI-SDR [3] averaged over organiser-synthesised clips plus $100 \cdot \Delta\text{WER}$, the word-error-rate reduction over all clips.

Challenges and research gaps. (i) *Low local SNR and phase:* under 56% of the labelled noise spans speech is present at a median local SNR of -7.3 dB; magnitude masks keep the noisy phase [4] and reach only 17.7 dB SI-SDR even as oracles. (ii) *Two objectives, one fixed recogniser:* enhancement artefacts can hurt recognition even when SNR improves [5], so fidelity and recognisability must be balanced for a recogniser we cannot adapt. (iii) *Scarce, shifting in-domain data:* only 695 paired validation clips exist, and the Phase 2 generator differs from Phase 1 (Table I). (iv) *A finite noise pool:* synthetic clips reuse recordings from a pool that the validation release exposes, which turns part of the task into known-interference cancellation [6], but the copies are slightly altered.

Contributions. (1) An event-conditioned complex-mask TF-DPCRN that takes noise-event windows from our Track 1 detector [7] as an input channel; (2) training data that reproduces measured test statistics, with staged fine-tuning and checkpoint selection grounded on organiser data; (3) a hybrid of exact known-noise subtraction and learned residual cleanup; (4) a recogniser-oriented continuation with a heavier log-mel loss. *Final standing:* **1st place**, Combined **26.4894** (SI-SDR 20.34 dB; ΔWER 6.15 points, 18.09 on synthetic and 0.52 on natural clips; transcript audit passed), ahead of the second-ranked entry (21.24).

II. Related Work

Deep noise suppression has moved from magnitude masking to complex-domain models such as DCCRN [8] and DPCRN [9], whose dual-path blocks [10] model frequency and time separately, trained with scale-invariant objectives [3] on large synthetic corpora [11]; complex ratio masks [12] recover phase. Enhancement for recognition must avoid the artefacts a downstream recogniser is sensitive to [5]. Classical adaptive noise cancelling subtracts interference given a reference signal [6]. We combine both views: references from a known pool where available, and a learned, event-conditioned model for the residual and for unknown noise.

III. Training Data and Preprocessing

A. Vaani data

Speech beds come from Vaani [1], [13]: 14,562 utterances (21.0 h, 35 districts) selected by duration (2–16 s), estimated SNR, clipping and voice activity (WebRTC VAD [14]). From the gold tier of the Vaani Noise Event Dataset [15] we take 459 noise events, 2,900 event-free clean segments (2.9 h) and 75 noise floors. Audio is 16 kHz mono, in 6 s (base) or 8 s (fine-tuning) crops.

B. Official validation release

Each of the 695 synthetic validation clips ships its clean reference, so *noise* = *mixture* $-$ *clean*; mixing is additive and confined to the labelled spans. Under 30.3% of the spans the reference is digital silence (the ideal output is silence); under 56.4% speech is present at -7.3 dB median local SNR. A

TABLE I
SYNTHETIC TEST CLIPS OF THE TWO PHASES, MEASURED WITH THE NOISE BANK ON 100 CONFIDENTLY SYNTHETIC CLIPS PER PHASE.

	Phase 1	Phase 2
Median clip length (s)	6.1	12.7
Noise events per clip	3.7	6.0
Events over digital silence	25.5%	10.1%
Local SNR of events over speech, median (dB)	-7.6	-9.7
Residual after subtracting the matched copy (dB)	-58	-26

file-name-group split gives a training fold (557 pairs; 0.79 h of pairs, 2,164 noise bursts, 530 floors) and a held-out fold of 138 pairs whose clean speech is never trained on; 374 further floors are harvested from natural validation clips. No other data is used.

C. Test-matched mixture generation

Base mixer: with probability 0.65 an organiser training pair (random crop, gain ± 6 dB, noise remixed within the fold); otherwise a Vaani or gold bed with a noise floor and 0–4 events whose count, duration and local SNR follow distributions measured on the organiser pairs (widened by 35%), with reverberation and channel EQ. *Phase 2 mixer:* measured with the noise bank on 100 confidently synthetic test clips, the Phase 2 generator places fewer events over silence and louder noise under speech, and alters the noise copies (Table I); the fine-tuning mixer reproduces these statistics (0.47 events/s, local SNR $\mathcal{N}(-9.7, 4)$ dB, 10% over silence, -22.8 dBFS) and alters each copy by a random 3-tap FIR, $\pm 0.4\%$ resampling and -35 dB hiss. Event windows are randomly dropped, jittered (up to 80 ms) or added (20%) so that the model tolerates detector errors.

IV. PROPOSED METHOD

A. Problem formulation

A synthetic test clip is $y = s + \sum_k g_k \, \tilde{n}_k(t - \tau_k)$, where each $\tilde{n}_k$ is a slightly altered copy of a pool recording n_k; natural clips contain unknown noise. We estimate the speech as

$$\hat{s} = f_\theta\Big(y - \textstyle\sum_k \hat{g}_k \, n_k(t - \hat{\tau}_k),\ w\Big) : \tag{1}$$

known copies are subtracted first (Sec. IV-D), and an enhancer f_θ conditioned on noise-event windows w removes the residual and all unknown noise.

B. Event-conditioned TF-DPCRN

Each member (3.9 M parameters, trained from scratch) takes the STFT X (512-point window, hop 128) as four channels: the real and imaginary parts of the power-compressed spectrum $X|X|^{-0.7}$ [16], the magnitude $|X|^{0.3}$, and a per-frame noise-event indicator w from the Track 1 windows. Four strided convolutions (48–128 channels) feed five dual-path blocks [10], [9], a frequency and a time BiLSTM with residual connections followed by self-attention over time [17], and a transposed-convolution decoder with skips outputs a complex ratio mask [12], [8] with bounded magnitude, $\hat{S} = \tanh(|M|)\, e^{j \angle M} X$.

TABLE II
TRAINING STAGES AND HELD-OUT SI-SDR (DB) OF THE 3-MEMBER ENSEMBLE. OFF0: 138 HELD-OUT ORGANISER PAIRS (INPUT 0.47 DB); P2H: 120 PHASE 2-LIKE MIXTURES (INPUT -5.2 DB). SELECTION: OFF0, P2H, OR FIT $= 1.53\,\text{OFF0} + 0.27\,\text{P2H}$.

Stage	Data	min	lr	Sel.	off0	p2h
base	base mixer + polish	185+25	10^{-3}	off0	23.3	13.3
p2ft	Phase 2 mixer	40	$2 \cdot 10^{-4}$	p2h	23.0	13.9
cont	Phase 2 mixer	45	10^{-4}	p2h	23.0	14.0
long	Phase 2 mixer	300	10^{-4}	p2h	23.5	14.7
p2mix	+ organiser pairs	240	10^{-4}	fit	24.1	14.9
asr	+ log-mel weight 1.2	40	10^{-4}	fit	24.2	14.9

C. Objective and staged training

$$\mathcal{L} = -\text{SI-SDR}(\hat{s}, s) + 0.1\, \mathcal{L}_{\text{MR}} + \lambda_{\text{mel}}\, \mathcal{L}_{\text{mel}}, \tag{2}$$

where $\mathcal{L}_{\text{MR}}$ averages spectral-convergence and log-magnitude losses [18] at three STFT resolutions and $\mathcal{L}_{\text{mel}}$ is an L1 log-mel distance (128 bands) aimed at the recogniser ($\lambda_{\text{mel}} = 0.3$, raised to 1.2 in the last stage). AdamW [19], warm-up and cosine decay [20], bf16 and weight averaging are used throughout; each stage ran on one 71 GB MIG slice of an NVIDIA H200. Table II lists the stages: base training on the base mixer, three fine-tuning stages on the Phase 2 mixer, one with real organiser pairs in the batches (30–50%), and the recogniser-oriented continuation.

D. Known-noise subtraction (disclosure)

In the validation release, *mixture − clean* at the labelled timestamps gives 2,213 events, i.e. 1,181 distinct recordings. On clips that the Track 1 routers call synthetic (1,635 of 3,178), each clip is pre-emphasised, resampled to 8 kHz and cross-correlated with every unit-norm template by FFT, normalised by the local signal energy [21]; matches are picked greedily by explained energy. Each match with correlation ≥ 0.45 is refined by ± 32 samples at 16 kHz and subtracted with a non-negative least-squares gain (9,065 copies). Phase 2 copies are altered by about -26 dB relative to the originals, so a residual remains; the enhancer therefore runs again over the subtracted spans (± 0.1 s) and the Track 1 windows, leaving the rest of the clip untouched. The organisers confirmed in writing that noise located in test audio by cross-correlation with validation-derived noise waveforms may be subtracted in Track 2, provided the method is disclosed, as done here.

E. Inference and implementation

The three members' outputs are averaged after aligning their polarity with the mixture, blended with the input outside the event windows ($y = \alpha\hat{s} + (1 - \alpha)x$, $\alpha = 0.9$), rescaled to the input level and transcribed by the unchanged SraVaani-1.0 (ARTPARK-IISc/SraVaani-1.0, revision f5dd535), the only pretrained model involved. With PyTorch 2.11 [22], inference takes about 40 min on one MIG slice. Code, configurations, all three candidate ensembles and the noise bank: https://github.com/sajidjavid222/CodeAMU_Track2.

TABLE III
PHASE 2 LEADERBOARD (3,178 CLIPS) AS THE SYSTEM WAS BUILT UP. SI-SDR IN DB ON SYNTHETIC CLIPS; ΔWER IN POINTS OVER ALL CLIPS; "–": NOT RECORDED.

System	SI-SDR	ΔWER	Comb.
Base ensemble (Phase 1 recipe)	12.13	1.43	13.56
+ test-matched fine-tuning (3 stages)	12.92	2.28	15.20
+ organiser pairs (p2mix)	–	–	15.64
+ log-mel continuation (asr)	13.38	2.29	15.67
Subtraction, no cleanup pass	–	–	19.53
+ cleanup, ASR-encoder-loss members	20.10	6.16	26.27
+ cleanup, p2mix	–	–	26.44
+ cleanup, p2mix and asr (5 models)	–	–	26.46
+ cleanup, asr (final, 1st)	20.34	6.15	**26.49**

V. RESULTS AND CONCLUSION

A. Protocol and main results

Models were selected on held-out organiser pairs (off0) and Phase 2-like mixtures (p2h), with the leaderboard as the final check. Table III decomposes the final system: the learned enhancer alone reaches 15.67, subtraction without the cleanup pass 19.53, and their combination 26.49, a gain of 5.25 over the second-ranked entry.

B. Analysis

Adapting to the shifted generator. The base ensemble scored 17.46 dB SI-SDR on the Phase 1 test but 12.13 dB on Phase 2. Test-matched fine-tuning recovered +1.64 Combined and real organiser pairs +0.44. Across our adapted submissions the leaderboard followed $-24.68{+}1.53\,\mathrm{off0}{+}0.27\,\mathrm{p2h}$ (maximum residual 0.12): the organiser pairs predict the test 5.7 times more strongly than our own mixtures, and an ensemble that improved only on the latter lost 0.58, so checkpoint selection was grounded on organiser data.

Subtraction and residual cleanup. With exact copies (Phase 1) subtraction alone scored 53.24; the Phase 2 alteration cut it to 19.53. Cleaning the residual restores most of the gap: in simulation on held-out pairs with -26 dB alterations, SI-SDR rises from 26.5 dB after subtraction to 31.4 dB after the cleanup pass.

Recognisability. Members trained with a loss on SraVaani's encoder features improved model-only submissions (16.18 vs 15.64), but on the nearly clean subtracted audio they cost SI-SDR (20.10 vs 20.34 dB, Combined 26.27); the milder log-mel continuation balanced both objectives and gave the final 26.49.

C. Limitations and future work

Subtraction applies only where the noise comes from a pool that the validation release exposes; elsewhere the learned enhancer alone applies (15.67). Gains on natural recordings remain small (ΔWER 0.52 points), where the noise is unknown and no reference exists. Residual models conditioned on the matched templates, and recogniser-aware objectives that do not cost fidelity, are natural next steps.

D. Conclusion

Combining known-noise subtraction with an event-conditioned complex-mask enhancer adapted to the measured test distribution gave the winning Track 2 system (1st place, Combined 26.4894). The released pipeline regenerates each bank-based submission exactly: all 3,178 WAV files bit-identical and all transcripts identical.

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